EXERCISE 1
SQL FUNDAMENTALS

1. SELECT * FROM employees;

2. SELECT DISTINCT department
FROM employees;

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department
IT
HR
finance
Marketing

3. SELECT first_name, last_name
FROM employees
ORDER BY salary DESC;

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first_name	last_name	salary
Emily	Davis	62 000
Mike	Johnson	60 000
Robert	Wilson	59 000
John	Doe	55 000
Daniel	Clark	53 000
Sarah	Brown	53 000
David	White	52 000
Jessica	Moore	51 000
laura	Hall	50 000
Jane	Smith	45 000

~~4. SELECT first_name,~~
~~last_name,~~
~~Sary Salary AS highest_Paid~~
~~FROM employees~~
~~LIMIT 5;~~

4. SELECT first_name, last_name, salary AS highest_paid
FROM employees
ORDER BY Salary DESC
LIMIT 5;

↓

first_name	last_name	Highest Paid
Emily	Davis	62 000
Mike	Johnson	60 000
Robert	Wilson	59 000
John	Doe	55 000
Daniel	Clark	53 000

5. SELECT first_name, last_name
FROM employees
WHERE department = 'IT';

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first_name	last_name	department
John	Doe	IT
Sarah	Brown	IT
Emily	Davis	IT
Laura	Hall	IT

6. SELECT first_name, last_name
FROM employees
WHERE department = 'finance' AND Salary > 58 000;

first_name	last_name	Salary	department
Mike	Johnson	60 000	finance
Robert	Wilson	59 000	finance

7. SELECT first_name, last_name
FROM employees
WHERE department = 'HR' OR department = 'Marketing';

first_name	last_name	department
Jane	Smith	HR
David	White	Marketing
Jessica	Moore	HR
Daniel	Clark	Marketing

8. SELECT first_name, last_name
FROM employees
WHERE NOT department = 'IT';

first_name	last_name	department
Jane	Smith	HR
Mike	Johnson	finance
David	White	Marketing
Robert	Wilson	finance
Jessica	Moore	HR
Daniel	Clark	Marketing

9. SELECT first_name, last_name
FROM employees
WHERE department IN ('HR', 'IT', 'finance');

first_name	last_name	department
John	Doe	IT
Jane	Smith	HR
Mike	Johnson	finance
Sarah	Brown	IT
Emily	Davis	IT
Robert	Wilson	finance
Jessica	Moore	HR
Laura	Hall	IT

10. SELECT first_name, last_name
FROM employees
WHERE department='IT' AN salary>50 000 AN City='New York';

first_name	last_name	department	salary	city
John	Doe	~~55 01~~ IT	55 000	New York
Sarah	Brown	IT	53 000	New York

11. SELECT first_name, last_name
FROM employees
WHERE (department='finance' OR department='Marketing')
AND salary > 52 000
ORDER BY Salary DESC;

first_name	last_name	department	salary
Mike	Johnson	finance	60 000
Robert	Wilson	finance	59 000
Daniel	Clark	Marketing	53 000

12. SELECT DISTINCT city
FROM employees
WHERE department NOT IN ('IT', 'HR');

City
Los Angeles
San Francisco
Houston
Chicago

13. SELECT first_name, last_name
FROM employees
WHERE department <> 'finance' AND salary > 50 000
ORDER BY hire_date ASC;

first_name	last_name	department	salary	hire_date
Emily	Davis	IT	62 000	2015-02-14
David	White	Marketing	52 000	2016-04-10
John	Doe	IT	55 000	2018-06-15
Jessica	Moore	HR	51 000	2018-05-22
Sarah	Brown	IT	53 000	2021-03-25
Daniel	Clark	Marketing	53 000	2022-06-01

14. SELECT first_name, last_name
FROM employees
WHERE (city = 'Chicago' OR city = 'Los Angeles')
AND department IN ('IT', 'Marketing')
LIMIT 3;

first_name	last_name	department	city
Emily	Davis	IT	Chicago
Daniel	Clark	Marketing	Chicago